

HDOC Day-36 – String Operations Using Functions

1. Algorithm

Step 1: Start the program.

Step 2: Display the menu:

1. Convert lowercase string to uppercase.
2. Reverse a string.
3. Check whether a string is a palindrome.
4. Exit.

Step 3: Read the user's choice.

Step 4: Use a `switch` statement:

- **Choice 1:** Call `toUppercase()`. Read a string, check each character, and convert lowercase letters to uppercase using ASCII values.
- **Choice 2:** Call `reverseString()`. Read a string, find its length manually, and print characters from the last character to the first.
- **Choice 3:** Call `checkPalindrome()`. Compare characters from the beginning and end of the string. If all corresponding characters match, it is a palindrome.
- **Choice 4:** Exit the program.
- **Default:** Display "Invalid choice".

Step 5: Repeat the menu until the user selects Exit.

Step 6: Stop.

2. C Program

```
#include <stdio.h>

void toUppercase()
{
    char str[100];
    int i;

    printf("Enter a lowercase string: ");
    scanf("%s", str);

    for (i = 0; str[i] != '\0'; i++)
    {
        if (str[i] >= 'a' && str[i] <= 'z')
        {
```

```

        str[i] = str[i] - 32;
    }
}

printf("Uppercase string: %s\n", str);
}

void reverseString()
{
    char str[100];
    int i, length = 0;

    printf("Enter a string: ");
    scanf(" %[^\\n]", str);

    while (str[length] != '\\0')
    {
        length++;
    }

    printf("Reversed string: ");

    for (i = length - 1; i >= 0; i--)
    {
        printf("%c", str[i]);
    }

    printf("\\n");
}

void checkPalindrome()
{
    char str[100];
    int i, length = 0;
    int flag = 1;

    printf("Enter a string: ");
    scanf(" %[^\\n]", str);

    while (str[length] != '\\0')
    {
        length++;
    }

    for (i = 0; i < length / 2; i++)
    {
        if (str[i] != str[length - 1 - i])
        {
            flag = 0;
            break;
        }
    }

    if (flag == 1)
        printf("The string is a palindrome.\\n");
    else
        printf("The string is not a palindrome.\\n");
}

```

```
}

int main()
{
    int choice;

    do
    {
        printf("\n----- STRING OPERATIONS -----\n");
        printf("1. Convert lowercase string to uppercase\n");
        printf("2. Reverse a string\n");
        printf("3. Check palindrome\n");
        printf("4. Exit\n");

        printf("Enter your choice: ");
        scanf("%d", &choice);

        switch (choice)
        {
            case 1:
                toUppercase();
                break;

            case 2:
                reverseString();
                break;

            case 3:
                checkPalindrome();
                break;

            case 4:
                printf("Exiting program...\n");
                break;

            default:
                printf("Invalid choice. Please enter 1 to 4.\n");
        }

    } while (choice != 4);

    return 0;
}
```

3. Test Case Table

Test Case	Choice	Input	Expected Output
TC-1	1	hello world	HELLO WORLD
TC-2	1	programming	PROGRAMMING
TC-3	2	hello	olleh
TC-4	2	computer	retupmoc
TC-5	3	madam	String is a palindrome
TC-6	3	level	String is a palindrome
TC-7	3	hello	String is not a palindrome
TC-8	4	—	Program exits
TC-9	7	—	Invalid choice

4. Test Case Evaluation

TC-1: The program checks every character of `hello world`. Each lowercase alphabet is converted to uppercase by subtracting 32 from its ASCII value. The output is `HELLO WORLD`. **Test Passed.**

```
——— STRING OPERATIONS ———
1. Convert lowercase string to uppercase
2. Reverse a string
3. Check palindrome
4. Exit
Enter your choice: 1
Enter a lowercase string: hello world
Uppercase string: HELLO WORLD
```

TC-3: The program manually calculates the length of `hello` and prints its characters from the last index to the first. The result is `olleh`. **Test Passed.**

```
——— STRING OPERATIONS ———
1. Convert lowercase string to uppercase
2. Reverse a string
3. Check palindrome
4. Exit
Enter your choice: 2
Enter a string: hello
Reversed string: olleh
```

TC-5: For `madam`, the program compares `m` with `m`, then `a` with `a`. Since all opposite characters match, it correctly identifies the string as a palindrome. **Test Passed.**

```
—— STRING OPERATIONS ——  
1. Convert lowercase string to uppercase  
2. Reverse a string  
3. Check palindrome  
4. Exit  
Enter your choice: 3  
Enter a string: madam  
The string is a palindrome.
```

TC-7: For `hello`, the first and last characters (`h` and `o`) do not match. Therefore, the program correctly reports that the string is not a palindrome. **Test Passed.**

```
—— STRING OPERATIONS ——  
1. Convert lowercase string to uppercase  
2. Reverse a string  
3. Check palindrome  
4. Exit  
Enter your choice: hello  
Enter a string: The string is not a palindrome.
```

TC-9: Since `7` is outside the available menu choices, the `default` case executes and displays an invalid-choice message. **Test Passed.**

```
—— STRING OPERATIONS ——  
1. Convert lowercase string to uppercase  
2. Reverse a string  
3. Check palindrome  
4. Exit  
Enter your choice: 7  
Invalid choice. Please enter 1 to 4.
```